

Hybrid Optical/Radio Frequency Communication Channel Model Final Reports

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Project Area: **Hybrid Optical/Radio Frequency Communication
Channel Model**

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Abstract

Hybrid RF/FSO communication systems offer a promising solution for reliable and high-speed data transmission. While RF signals are robust under poor visibility, FSO signals offer higher bandwidth in clear conditions. However, both channels are susceptible to weather-induced attenuation, requiring accurate prediction models for reliable system design. This study develops and compares Random Forest-based models to predict RF and FSO attenuation using synthetic data across seven weather conditions.

Three modelling strategies were explored: (1) specific models trained per weather type, (2) a generic model using all data with weather encoded as a categorical feature, and (3) joint models where RF predictions assist FSO predictions and vice versa. Models were evaluated using RMSE and R^2 , and further interpreted using 2D heatmaps, Pearson correlation, and entropy-based metrics.

Results showed that specific models consistently outperformed the generic model, especially in adverse conditions like fog and rain. After hyperparameter tuning, these models provided the most stable and accurate performance and were used as the baseline for joint models. Joint models captured additional cross-channel dependencies, but their performance varied across conditions.

Visual and statistical analyses confirmed the tuned specific model's reliability, interpretability, and robustness, making it the preferred approach for hybrid RF/FSO system prediction and a strong foundation for joint modelling.

1 Introduction

The rapid growth in data demand for modern communication networks has created quite challenges for existing communication systems. While radio frequency (RF) communication has been a foundation for wireless communication, its limited bandwidth and more likely to be interfered by poor weather conditions highlight the need for supporting technologies. For data transmission Free-space optical(FSO) uses infrared light, offers a promising solution with its high bandwidth and immunity to electromagnetic interference (ITU Radiocommunication Sector 2019). However, FSO systems are very sensitive to weather conditions such as rain, fog, and dust, which cause significant signal attenuation (Sector 2012).

Both the RF and FSO channels are affected by weather conditions, which can reduce communication quality. To address this, our study focuses on developing predictive models that can estimate signal attenuation based on weather parameters. We implement machine learning techniques, specifically Random Forest models, to model RF and FSO losses under different weather scenarios. The dataset used for this study is synthetic but derived from real-world hybrid systems, consisting of seven distinct weather conditions.

We undergo three modeling approaches: Specific Models: These models are trained for each weather condition, allowing them to capture unique patterns associated with specific weather conditions.

Generic Model: A single model trained using all weather conditions which aims to generalise the relationship between weather parameters and signal attenuation.

Joint Modeling Techniques: In this approach, predictions from one channel (RF or FSO) are used as input features to predict the attenuation in the other channel, capturing the interdependence between the two.

To further understand the uncertainties and dependencies between RF and FSO channels, we included measures such as entropy and mutual information. Entropy helps quantify the uncertainty in signal attenuation under different weather conditions, while mutual information assesses the degree of dependency between RF and FSO channels. These calculations and analysis provide deeper insights into the behaviour of hybrid communication systems and help in developing more resilient models.

By combining machine learning techniques with information-theoretic analyses, this study aims to enhance the predictive capabilities of hybrid RF/FSO communication systems, ensuring more reliable performance in the face of varying weather conditions.

2 Background

The hybrid communication system is the technology with the combination of RF and FSO which are emerging as a promising solution to address the limitations of single-mode communication systems. Both RF and FSO have unique characteristics that make them suitable for specific scenarios but also pose significant challenges, especially under varying atmospheric conditions.

2.1 Radio Frequency Communication

RF communication is a well built and used technology that forms the backbone of modern wireless networks, including 4G, 5G, and satellite-based systems. Its resilience to moderate weather conditions, such as rain and fog, makes it a reliable option for long-range and low-bandwidth communication (ITU Radiocommunication Sector 2019). However, RF channels face challenges such as limited bandwidth and susceptibility to electromagnetic interference, especially in dense urban environments. These limitations hinder its ability to meet the growing demand for high-data-rate applications.

2.2 Free-Space Optical Communication

FSO communication uses infrared light to transmit data through the atmosphere, offering several advantages over RF, such as higher bandwidth, immunity to electromagnetic interference, and low latency (Sector 2012; Chowdhury et al. 2020). It is useful for applications that require high-speed data transmission, such as data center interconnects, last-mile access, and satellite communication. However, FSO systems are highly sensitive to atmospheric disturbances, including rain, fog, dust, and turbulence, which significantly degrade signal quality. For instance, fog can cause signal attenuation as high as 30–40 dB/km, rendering FSO systems less reliable during adverse weather conditions (Han 2023).

2.3 Hybrid RF/FSO Systems

To mitigate or overcome the limitations of either RF or FSO systems, hybrid RF/FSO systems have been proposed. These systems combine the strengths of both channels: RF for reliable communication during poor visibility and FSO for high-speed data transmission during clear weather (Chowdhury et al. 2020; Nadeem et al. 2010). By dynamically switching between RF and FSO channels or using them simultaneously,

hybrid systems can maintain high performance across varying environmental conditions. Applications of hybrid RF/FSO systems include:

- **5G Backhaul and Fronthaul:** Addressing the bandwidth and latency requirements of next-generation wireless networks.
- **Satellite-Ground Communication:** By mitigating weather based disruptions, satellite communication can be enhanced.
- **Disaster Recovery:** During emergencies when conventional infrastructure is not available, provides robust communication.

2.4 Weather Impact on Hybrid Systems

Weather conditions are a critical factor in the performance of hybrid RF/FSO systems. Rain, snow, fog, and dust storms affect the attenuation levels of both RF and FSO channels. For example, RF signals experience rain attenuation due to scattering and absorption by raindrops, while FSO signals are heavily attenuated by fog and dust particles (ITU Radiocommunication Sector 2019; Sector 2012). Understanding the relationship between weather parameters and signal performance is crucial for designing effective hybrid systems.

2.5 Modeling Weather-Dependent Attenuation

Accurate modeling of weather-induced attenuation is vital for optimizing hybrid RF/FSO systems. Recent studies have proposed various methods, including empirical, statistical, and machine learning-based models, to predict attenuation levels based on weather metrics (Han 2023; Nadeem et al. 2010). Random Forest algorithms, in particular, have shown promise in handling highly nonlinear relationships and identifying key features influencing system performance. This project builds upon these advancements to develop predictive models that can inform the design and deployment of resilient hybrid communication systems.

3 Methods

This project focuses on developing predictive models for hybrid RF/FSO communication systems to analyze the impact of weather conditions on signal attenuation. The methodology is divided into five key stages: dataset preparation, feature engineering, model training, evaluation, and comparative analysis of specific and generic approaches.

3.1 Dataset Preparation

The dataset used in this project contains weather metrics and their corresponding attenuation values for RF and FSO channels. Key features include:

- **Weather Metrics:** Parameters such as rain intensity, visibility, temperature, and wind speed.
- **Categorical Variables:** `SYNOPCode`, which represents weather conditions like clear weather, rain, or dust storms, is handled as a categorical variable (ITU Radiocommunication Sector 2019; Sector 2012).
- **Target Variables:** Signal attenuation levels for RF (`RFL_Att`) and FSO (`FSO_Att`) channels.

To ensure consistent and accurate analysis, the required preprocessing steps are as follows:

- Handling missing values through imputation.
- Encoding categorical variables using one-hot encoding.
- Normalizing numerical features to improve model performance (Han 2023).

But since, the dataset is clean we don't need to use imputation, 'SYNOPCode' is a categorical one but its not needed to be one hot encoded and finally, for decision based trees like random forest, normalizing is meaningless. So, we can skip these steps.

3.2 Exploratory Data Analysis

The dataset underwent a thorough exploratory data analysis (EDA) to gain insights into variable distributions, relationships, and potential anomalies. The following steps were performed:

- **Descriptive Statistics:** Figure 1 shows summary statistics such as mean, median, standard deviation, and range were computed for all numerical variables.

Descriptive Statistics:

	FSO_Att	RFL_Att	AbsoluteHumidity	AbsoluteHumidityMax	AbsoluteHumidityMin	Distance	Frequency	Particulate	ParticulateMax	ParticulateMin	...
count	91379.000000	91379.000000	91379.000000	91379.000000	91379.000000	91379.000000	9.137900e+04	91379.000000	91379.000000	91379.000000	...
mean	6.769458	11.619098	9.553919	10.032760	9.076251	3297.930328	7.850005e+10	27.065979	28.417120	25.717089	...
std	3.903843	3.438873	5.858577	6.162798	5.575927	1224.305893	5.000027e+09	72.134023	75.761896	68.595239	...
min	0.788363	0.027142	1.141556	1.238270	1.049744	2012.000148	7.350000e+10	0.000000	0.000000	0.000000	...
25%	3.473063	10.829331	4.958993	5.205861	4.709511	2019.431812	7.350000e+10	0.000000	0.000000	0.000000	...
50%	6.336167	11.856560	6.870737	7.205499	6.524046	2959.863686	8.350000e+10	0.000000	0.000000	0.000000	...
75%	8.664984	12.847944	14.049470	14.782679	13.379256	4820.890157	8.350000e+10	16.947618	17.775980	16.038090	...
max	32.455222	46.893150	24.790883	26.407305	24.268431	4827.999971	8.350000e+10	1621.001906	1753.747866	1500.666382	...

8 rows x 27 columns

Figure 1: Summary Statistics for all the features

- **Distributions and Outliers:** Histograms were plotted to visualize feature distributions. Features such as rain intensity and visibility showed significant fluctuation, likely influenced by weather conditions as shown in Figure 2 and 3.

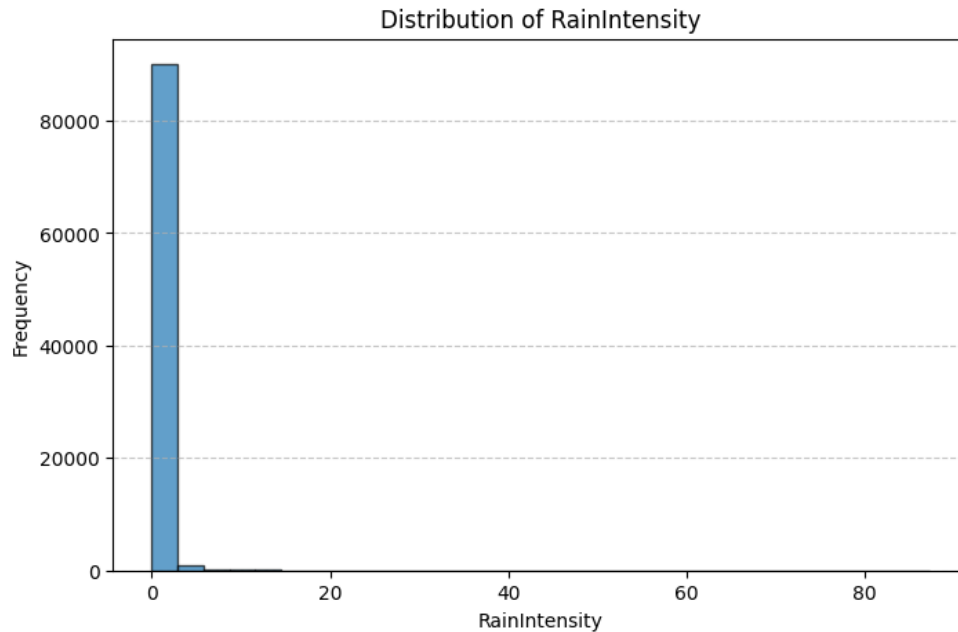


Figure 2: Histogram plot of Rain Intensity

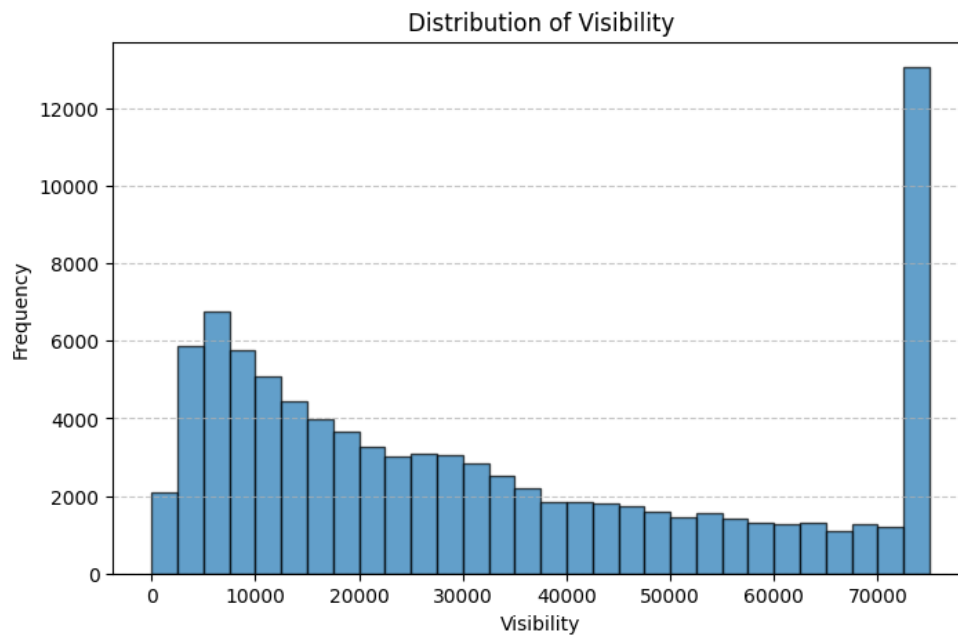


Figure 3: Histogram plot of Visibility

Figure 4 and 5 box-plots highlighted outliers in metrics such as temperature and particulate matter, which may impact model performance. `RFL_Att` exhibited slight skewness.

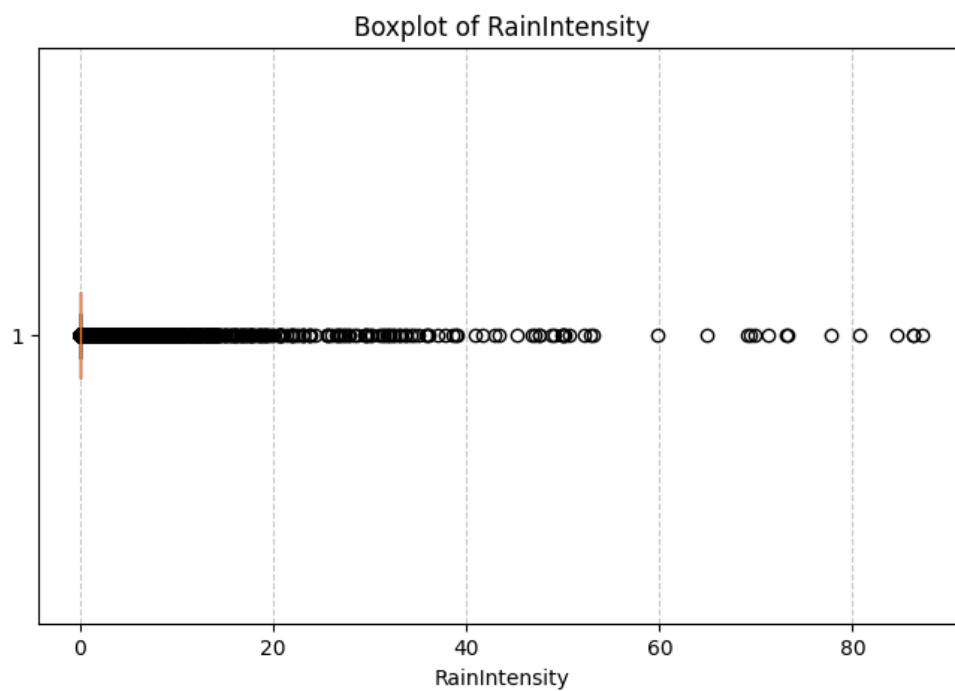


Figure 4: Box-plot plot of Rain intensity

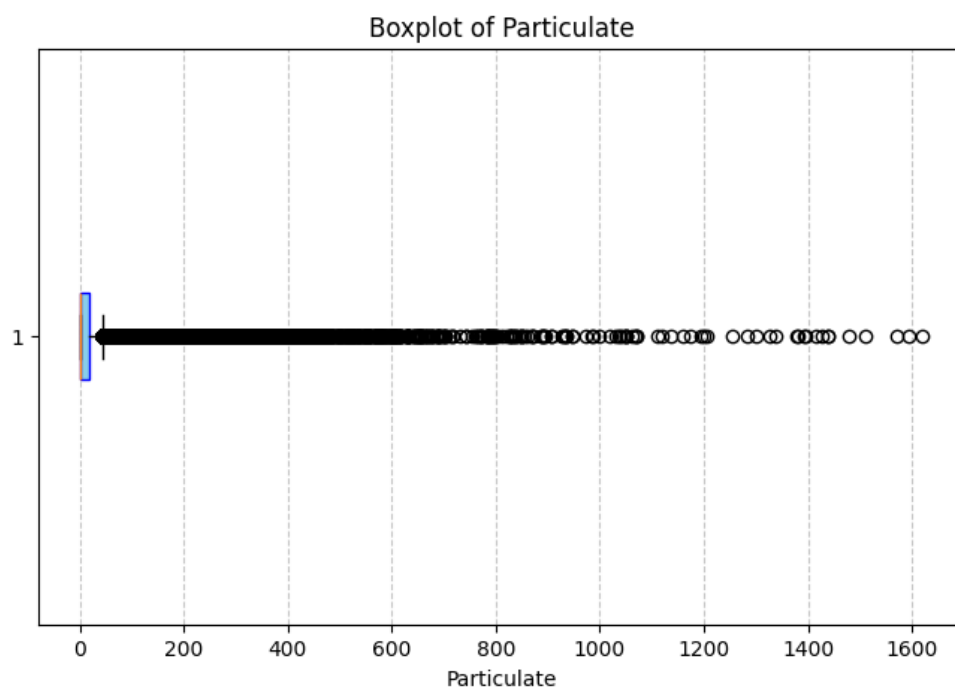


Figure 5: Box-plot plot of Particulate

- **Correlation Analysis:** The correlation matrix in Figure 6 revealed strong relationships between **FSO_Att** and features such as particulate matter and relative humidity. Similarly, **RFL_Att** was strongly correlated with rain intensity and particulate matter.

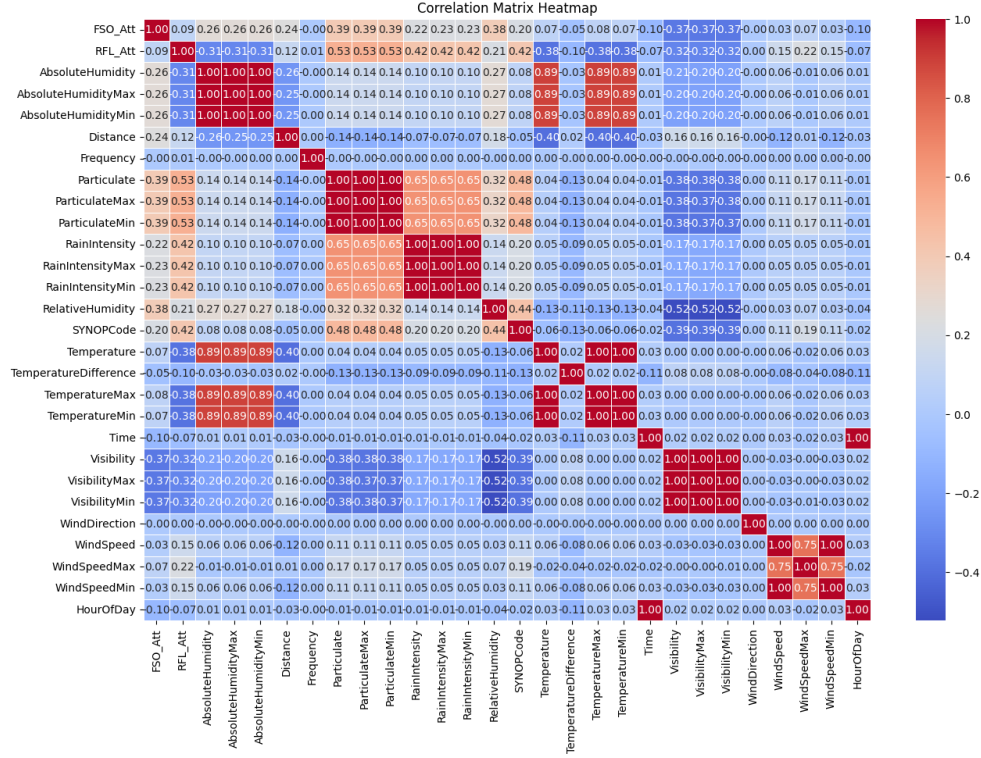


Figure 6: Heat map plot showing correlation of each feature

- **Trends:** Time-based analysis showed hourly trends in attenuation, with significant variations during adverse weather conditions (Chowdhury et al. 2020).

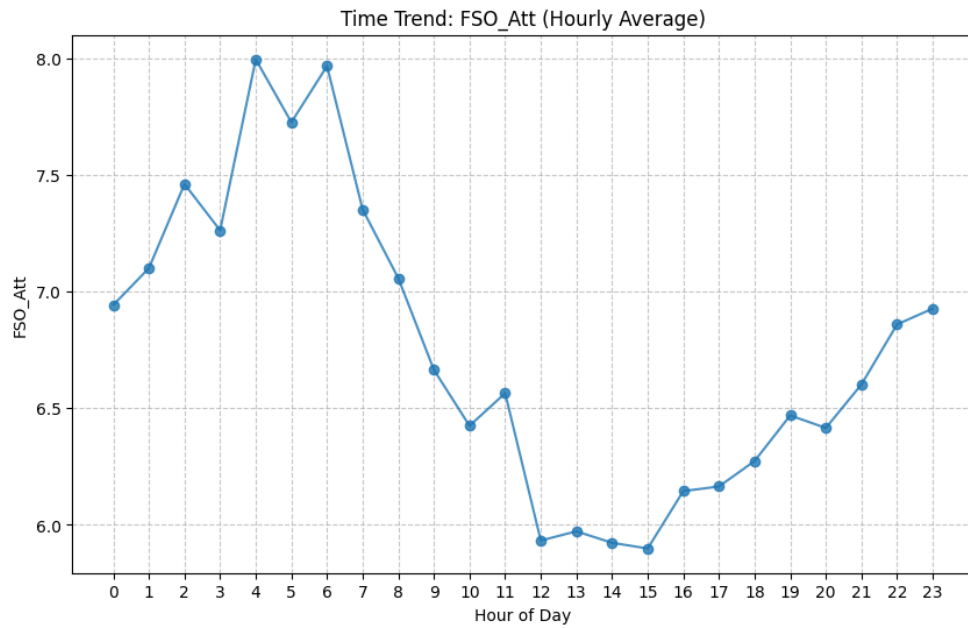


Figure 7: Time Trend for fso_att column

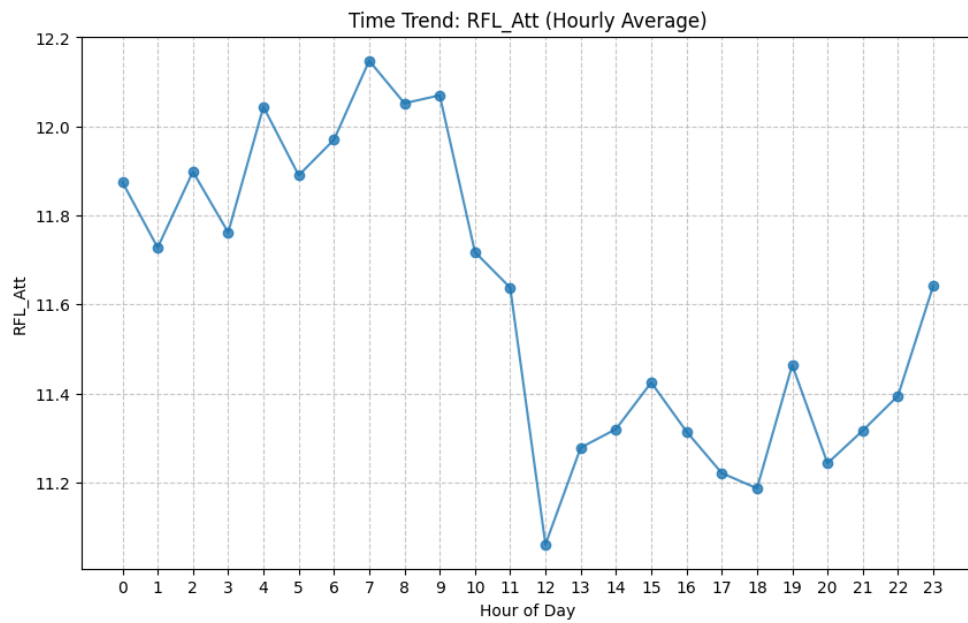


Figure 8: Time Trend for rfl_att column

Figures summarizing these analyses include histograms, boxplots for outliers, and the correlation heatmap.

- **Distribution of SYNOPCode:** Since, we will be training and testing the models based on SYNOPCode which is basically represents the 7 different weather conditions as shown in the Table 9.

clear,	dust storm,	fog,	drizzle,	rain,	snow	showers
0	3	4	5	6	7	8

Figure 9: Mapping of SYNOPCode
(Australian Weather News n.d.)

The Figure 10 shows the distribution of SYNOPCode on entire dataset which shows that dataset contains mostly of 'Clear' weather and least of 'Dust Storm' which implies that that the dataset is imbalanced. For the entire training and testing of the system, this distribution will be maintained as shown in Figure 11 and Figure 12. Moreover the training data contains 70

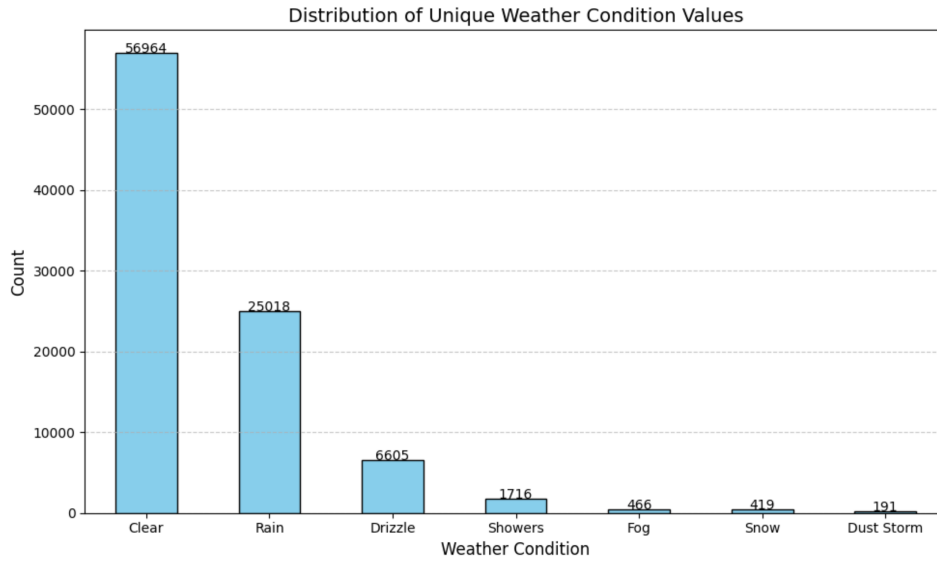


Figure 10: Distribution of SYNOPCode on whole dataset

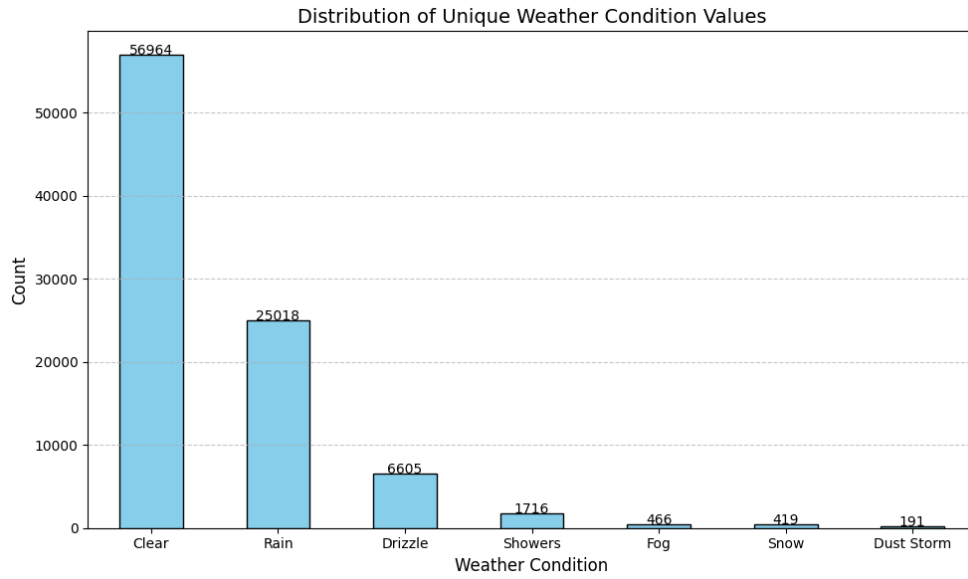


Figure 11: Distribution of SYNOPCode on train dataset

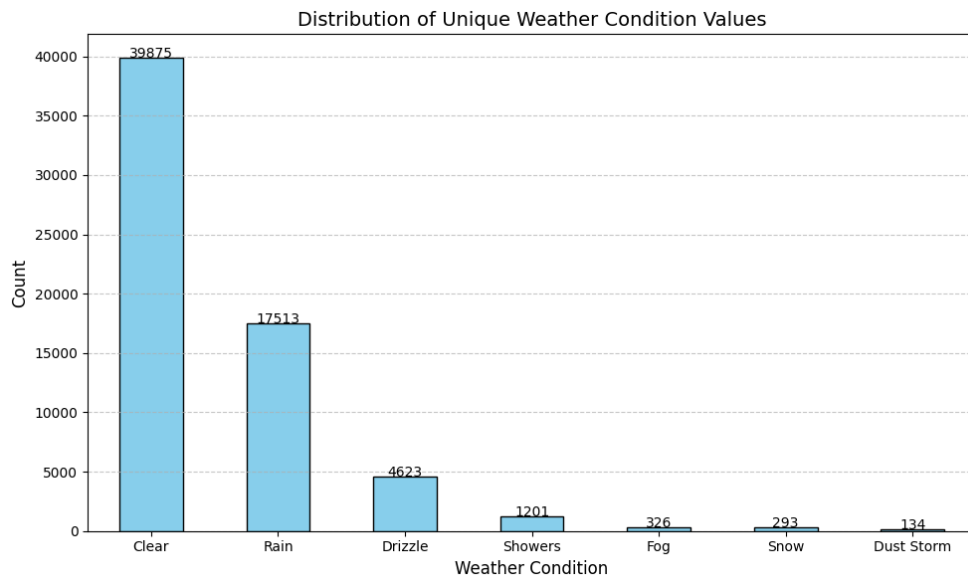


Figure 12: Distribution of SYNOPCode on test dataset

3.3 Feature Engineering

Feature selection was conducted to improve model performance and reduce redundancy:

- **Random Forest Importance:** Random Forest algorithms were used to rank features based on their contribution to predicting attenuation. Visibility feature was identified as the most critical for FSO attenuation, while absolute humidity was the primary factor for RFL attenuation.

Algorithm 1 An algorithm for ranking the importance of predictor variables.

- 1: Let $\mathcal{S}$ be the set of N predictor variables.
 - 2: Let $\mathcal{R}$ be an empty table.
 - 3: The training data containing only the variables in $\mathcal{S}$ are used to train a random forest.
 - 4: The RMSE and R^2 value for the random forest are calculated.
 - 5: The importance of predictor variables is ranked according to the out-of-bag information.
 - 6: The least important predictor is removed from $\mathcal{S}$ and is combined with the RMSE and R^2 value to form a new row at the end of the table $\mathcal{R}$.
 - 7: If $\mathcal{S}$ is non-empty, go to Step 3.
 - 8: The output is the table $\mathcal{R}$.
-

Figure 13: Feature Importance Ranking Algorithm

- **Dimensionality Reduction:** Features with minimal importance were removed iteratively, guided by model performance metrics such as RMSE and R^2 .

– Feature Selection for Generic Models:

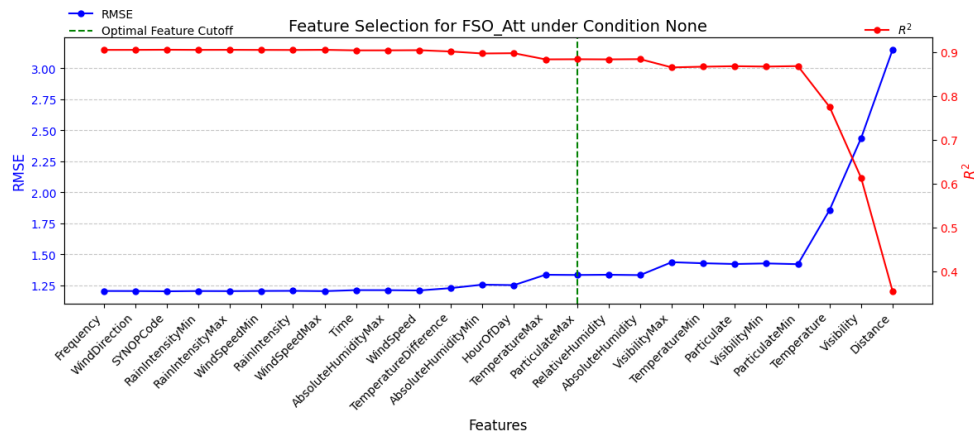


Figure 14: Feature Selection for Generic Model FSO Attenuation

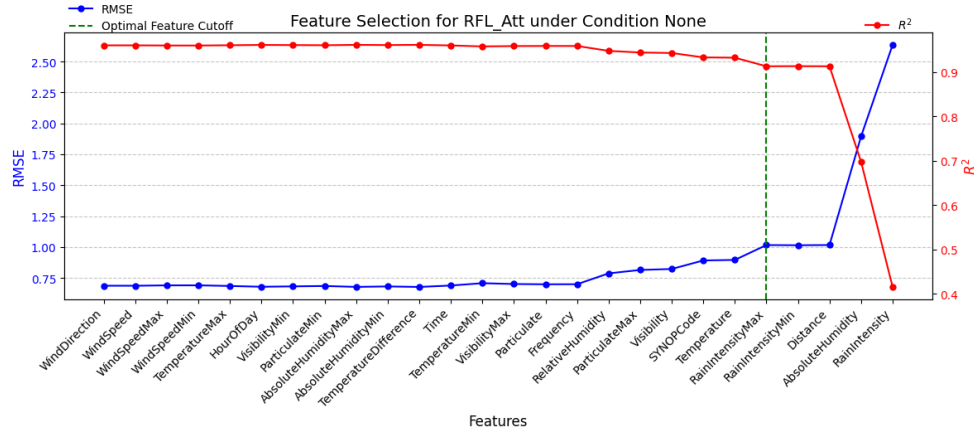


Figure 15: Feature Selection for Generic Model RFL Attenuation

Figure 14 and 15 shows the impact of each feature on the RF attenuation and FSO attenuation. The green dotted line is the line that is the cutoff point which is created dynamically by observing the changes in RMSE change is less than 5% and took the features from that point to the right as selected feature to train generic model for FSO and RFL Attenuation.

- **Feature Selection for Specific Models:** Likewise, Figure 16 and 17 show the impact of each feature on the RF attenuation and FSO attenuation for SYNOPCode '0' (Clear Weather). Then, selected features are used to train specific models for each SYNOP code making total of 7 models.

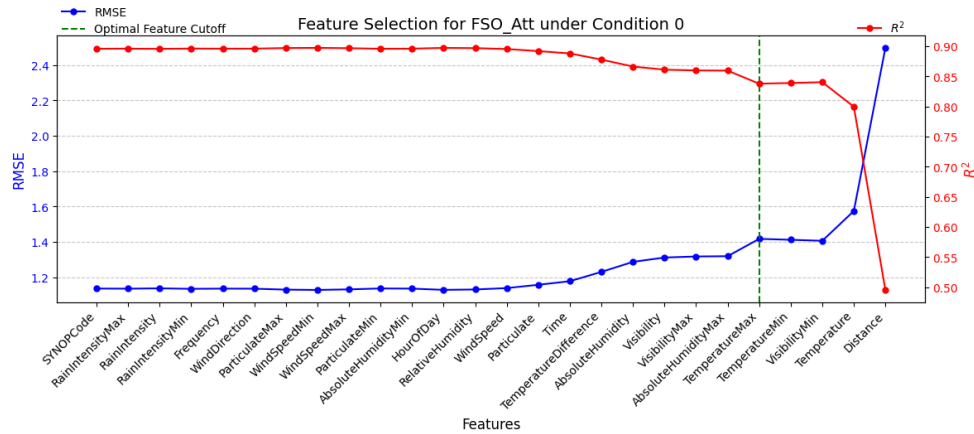


Figure 16: Feature Selection for Specific Model FSO Attenuation (Clear Weather)

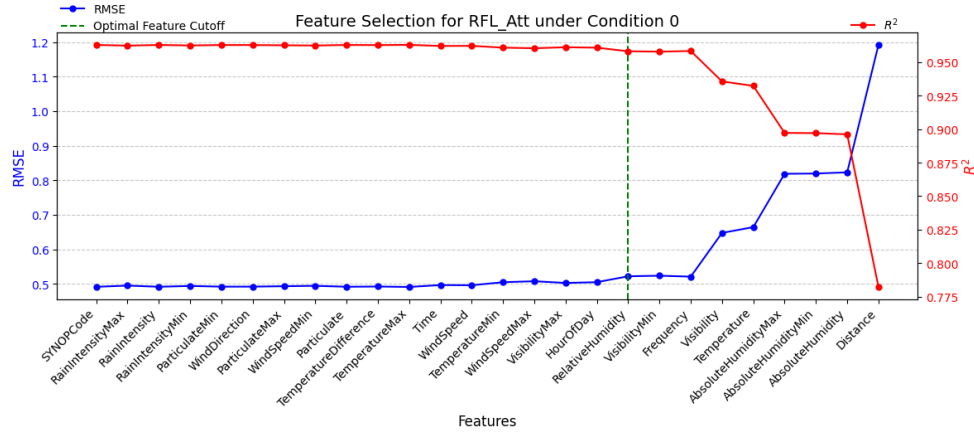


Figure 17: Feature Selection for Specific Model RFL Attenuation (Clear Weather)

- **Feature Selection for Joint Model:** For this model, we will be using best model (Specific model) to do the feature selection by getting RFL prediction and using it as a feature to predict FSO and vice versa as shown in Figure 18 and 19.

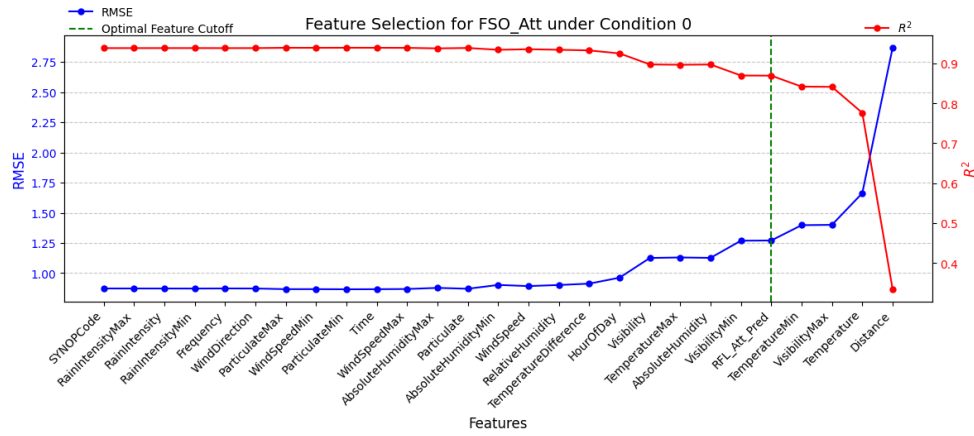


Figure 18: Feature Selection for Joint Model FSO Attenuation (Clear Weather)

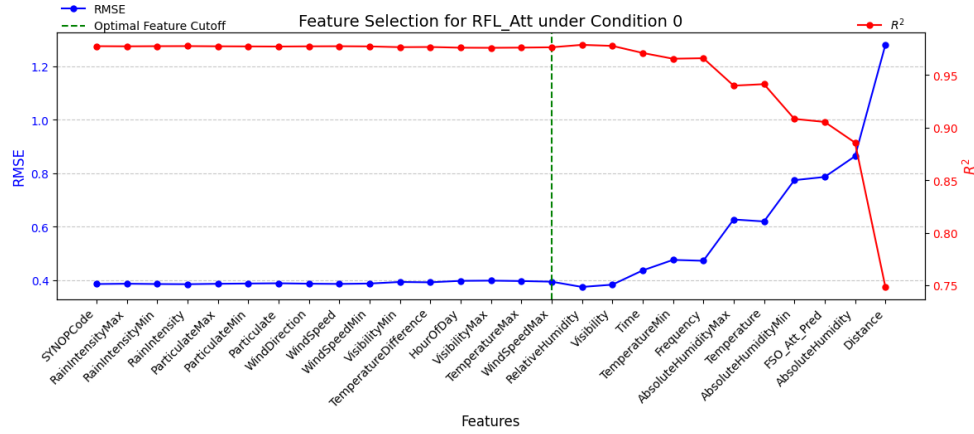


Figure 19: Feature Selection for Joint Model RFL Attenuation (Clear Weather)

- **Hyperparameter Tuning:** In this process, we set a set of hyperparameters prior to training to find best set of hyperparameters for random forest model.

Key hyperparameters for Random Forest models:

- **n_estimators:** The number of trees in the forest.
- **max_depth:** The maximum depth of each tree.
- **min_samples_split:** The minimum number of samples required to split an internal node.
- **min_samples_leaf:** The minimum number of samples required to be at a leaf node.
- **max_features:** The number of features to consider when looking for the best split.

(Muller 2020)

Hyperparameter tuning is performed in order to enhance model accuracy by finding optimal balance between bias and variance, prevent overfitting by managing the complexity of the model and help model from overfitting and improve generalisation to unseen data.

For this process, RandomizedSearchCV is used as compared to exhaustive methods like grid search, it is more efficient to deal with large number of hyperparameters and reduce long tuning process. The tuning process is done in following steps:

- Defining hyperparameter Space or grid

- Randomly sample the combinations of hyperparameters from defined hyperparameter grid
- Ideally evaluate each combination using the cross validation but for this dataset test set is used as the distribution of SYN-OPCode based dataset is very imbalanced.
- The combination giving best performance score will be the final model.

```
# Defining the hyperparameter grid for RandomForest tuning
param_grid = {
    'n_estimators': [50, 100, 150],
    'max_depth': [10, 30, None],
    'min_samples_split': [2, 4, 5]
}
```

Figure 20: Hyperparameter Grid

This grid is used to test a mix of model sizes and complexities. `n_estimators` controls the number of trees — 50 is quicker, 100 is standard, and 150 checks if more trees help. `max_depth` limits how deep each tree can grow — 10 keeps it simple, 30 adds more flexibility, and `None` lets it grow fully. `min_samples_split` prevents overfitting — using values 2, 4, and 5 helps us see if slightly stricter splits improve performance. This setup gives us a good balance between speed and accuracy.

3.4 Model Training

Three distinct modeling approaches were employed:

- **Specific Models:** Separate models were trained for each weather condition defined by `SYNOPCode`. This approach allowed the models to specialize in handling unique conditions like heavy rain (ITU Radiocommunication Sector 2019).
- **Generic Model:** A single model was trained using the entire dataset, with `SYNOPCode` included as a feature to generalize across all conditions (Chowdhury et al. 2020).
- **Joint Models:** Inorder to see the interdependence between RF and FSO channels, two joint modelling approaches were used:
 - **Method 2 (RFL → FSO) :** In this model, predicted RFL attenuation values are used as additional input feature to the FSO model. This approach used the information from RFL channel to enhance the prediction of FSO attenuation.

- **Method 2 (FSO $\rightarrow$ RFL)** : On the other hand, the predicted FSO attenuation values are used as one of the input feature to to the RFL model.

Random Forest regressors were used for both methods due to their robustness in handling nonlinear relationships, handle uneven (imbalanced) data and categorical variables (Han 2023). The dataset was split into training (70%) and testing (30%) subsets, ensuring no data leakage.

3.5 Evaluation

The models were evaluated using the following metrics:

3.5.1 Root Mean Square Error (RMSE)

The RMSE measures the average squared of prediction errors and is defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Where:

- y_i : Actual value of the target variable.
- $\hat{y}_i$: Predicted value.
- n : Number of observations.

A lower RMSE indicates better model performance, as it reflects smaller deviations between predicted and actual values (Sector 2012).

3.5.2 Coefficient of Determination (R^2)

The R^2 metric measures how well the model explains the variance in the target variable and is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Where:

- $\bar{y}$: Mean of the actual values.
- $\sum_{i=1}^n (y_i - \hat{y}_i)^2$: Residual sum of squares.
- $\sum_{i=1}^n (y_i - \bar{y})^2$: Total sum of squares.

An R^2 value closer to 1 indicates better performance, as it implies the model explains most of the variance in the target variable (Han 2023; Nadeem et al. 2010).

3.5.3 Pearson Correlation Coefficient (r)

The Pearson Correlation Coefficient measures the strength and direction of the linear relationship between two continuous variables. Its values range from -1 to $+1$:

- $+1$: perfect positive linear correlation
- 0 : no linear correlation
- -1 : perfect negative linear correlation

The formula is given by:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

where:

- x_i and y_i are the individual sample points,
- $\bar{x}$ and $\bar{y}$ are the means of the respective variables,
- n is the total number of observations.

This metric helped us understand how RFL and FSO attenuations were linearly related across different weather conditions.

3.5.4 Entropy (H)

Entropy measures the level of uncertainty or randomness in a variable, based on its probability distribution. Higher entropy indicates more unpredictability.

The entropy of a discrete variable X is given by:

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i)$$

where:

- $p(x_i)$ is the probability of occurrence of the i^{th} value,
- n is the number of distinct values the variable X can take.

In this study, entropy helped identify which weather conditions introduced more uncertainty in RFL and FSO attenuation.

3.5.5 Mutual Information ($I(X; Y)$)

Mutual Information quantifies the amount of information one variable contains about another, capturing both linear and non-linear relationships.

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 \left(\frac{p(x, y)}{p(x)p(y)} \right)$$

where:

- $p(x, y)$ is the joint probability distribution of X and Y ,
- $p(x)$ and $p(y)$ are the marginal distributions of X and Y respectively.

This metric allowed us to examine how strongly RFL and FSO attenuation were dependent on each other under various conditions.

4 Results and Discussion

This section presents the performance of the models developed using different approaches: specific models (per weather condition), a generic model (trained on all weather types), and joint models (where predicted RF/FSO is used as a feature). The models were evaluated using RMSE and R^2 , along with additional analysis using entropy, Pearson correlation, and mutual information. Moreover, the below discussion is just an observation of performance metrics rather than preference.

4.1 Model Performance

The performance of the models was first compared using the average RMSE and R^2 values across all weather conditions. As seen in Figure 21 and Figure 22, the specific model showed lower RMSE and slightly higher or comparable R^2 values than the generic model for both FSO and RF attenuation. This suggests that training a separate model for each weather condition allows the model to better capture condition-specific relationships between weather features and attenuation.

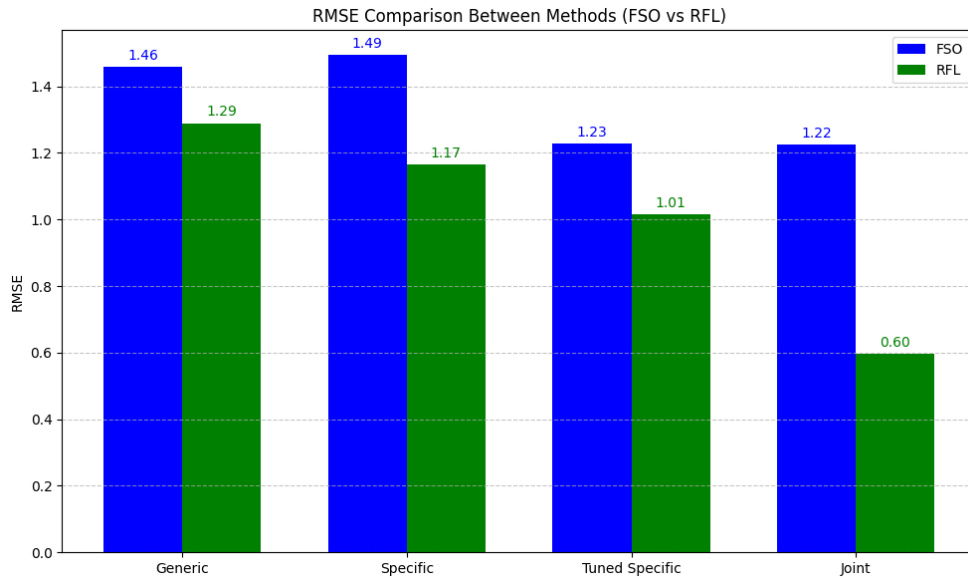
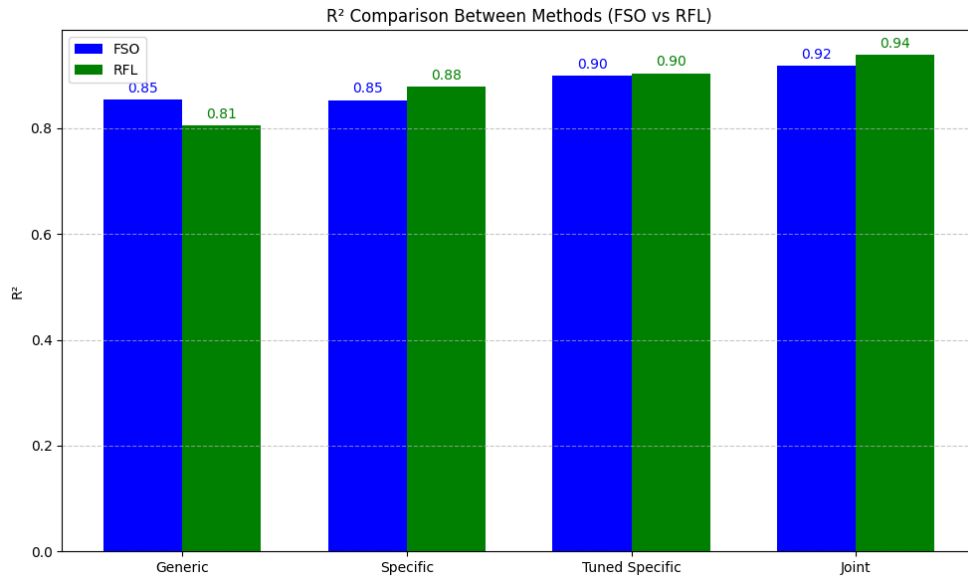


Figure 21: Average RMSE across all models

Figure 22: Average R^2 across all models

4.2 Performance Across Weather Conditions

To better understand how model performance varies across different weather types, RMSE and R^2 values were also analysed per weather condition. Figure 23 and Figure 24 present line plots comparing all four model types.

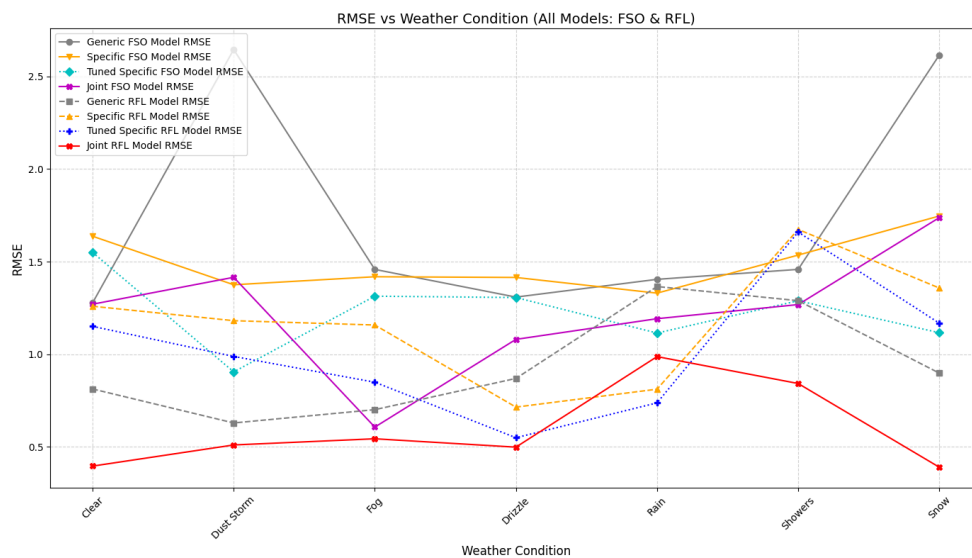


Figure 23: RMSE of all models based on weather conditions

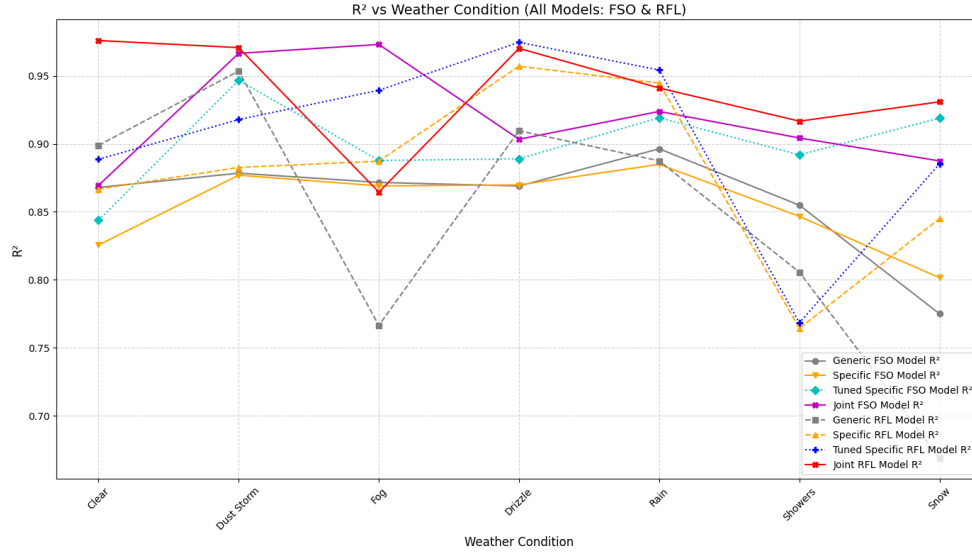


Figure 24: (R^2 of all models based on weather conditions

These plots show that the specific model consistently performed better or comparably under most weather types, with notably lower RMSE in fog, drizzle, and snow. This demonstrates that the specific model is more adaptable to condition-specific variations, especially in adverse weather conditions where generalisation from a global model (generic) is difficult.

4.3 Model Selection Justification

Given the superior performance of the specific models in both average and weather-specific evaluations, they were selected as the base model for further tuning. Hyperparameter tuning was applied to improve the specific model further. Once optimised, the tuned specific models achieved the lowest overall RMSE values across conditions.

Moreover, because specific models showed better ability to capture detailed condition-level behaviour, they were also used as the foundation for the joint model strategy. This ensured that the predictive benefit of including information from the complementary channel (i.e., RF helping FSO or vice versa) was built on a strong and weather-aware baseline model.

4.4 Visual Evaluation using Heatmaps

Adjusted Bin Size Rule for Heatmaps: To visualise the joint distribution of RF and FSO attenuation using 2D histograms or heatmaps,

we needed to determine a suitable number of bins. An excessively fine grid may lead to sparsity and noise, whereas overly coarse binning may obscure important patterns. We applied a modified version of the well-known square-root rule, often used in histogram binning:

$$\text{Number of bins per axis} = \left\lceil \frac{\sqrt{n}}{k} \right\rceil$$

where n is the number of data points, and k is a scaling factor. In this study, we chose $k = 5$ to reduce visual clutter and improve interpretability, especially for high-density regions. This adjustment ensures that the heatmap captures the overall structure without becoming too granular.

To visually inspect how well each model replicates the real attenuation behaviour under different weather conditions, 2D heatmaps of RF vs FSO attenuation were plotted. Each weather type includes four heatmaps: the actual data, the tuned specific model prediction, and both directions of the joint models (RF→FSO and FSO→RF).

The comparison across all weather types (Figures 25 to 27) shows that the tuned specific model preserves the high-density regions from the actual data quite well. The joint models also captured meaningful trends, particularly under fog and rain, where cross-channel dependencies are more apparent. However, under certain conditions like dust storms, the joint model struggled slightly due to sparse data distribution.

These visual patterns further reinforce the decision to use the tuned specific model as the base for joint modelling, as it provides stable and interpretable output even when generalisation is more difficult.

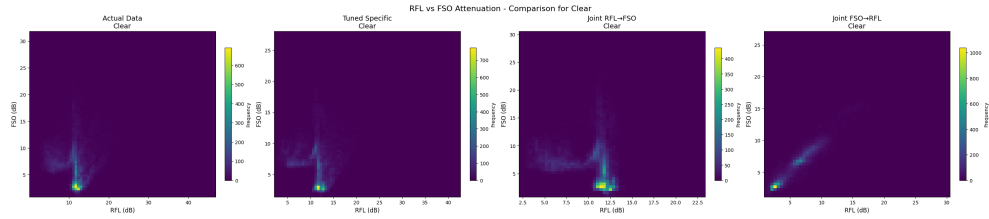


Figure 25: RF vs FSO Attenuation Heatmaps under Clear Weather

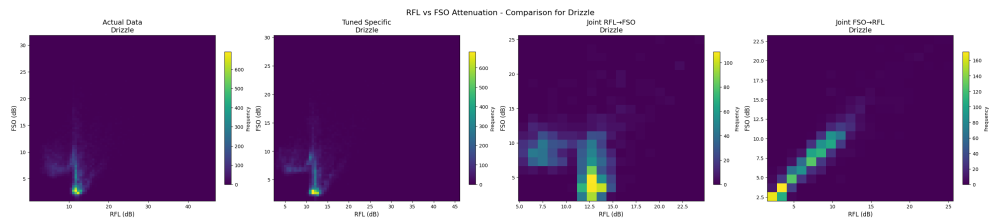


Figure 26: RF vs FSO Attenuation Heatmaps under Drizzle Weather

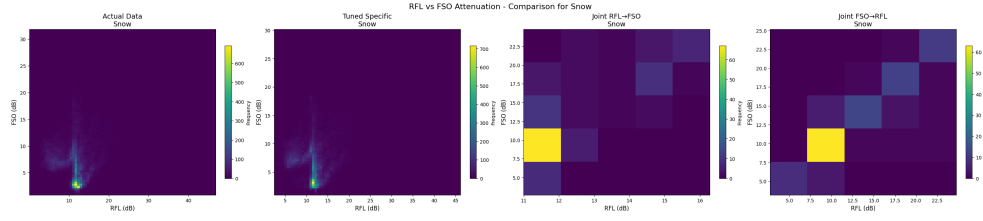


Figure 27: RF vs FSO Attenuation Heatmaps under Snow Weather

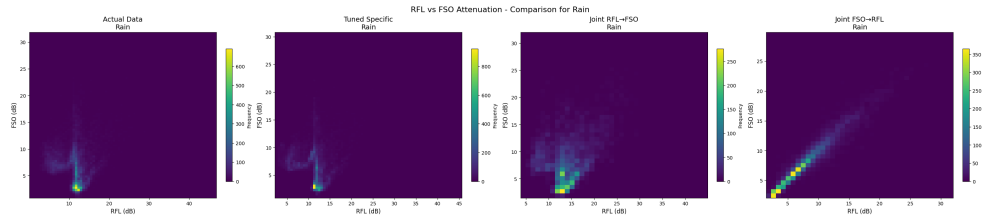


Figure 28: RF vs FSO Attenuation Heatmaps under Rain Weather

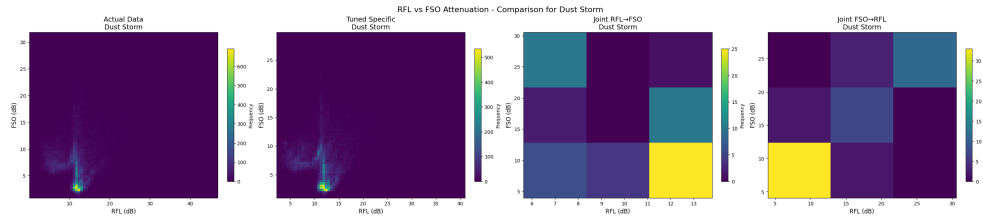


Figure 29: RF vs FSO Attenuation Heatmaps under DustStorm Weather

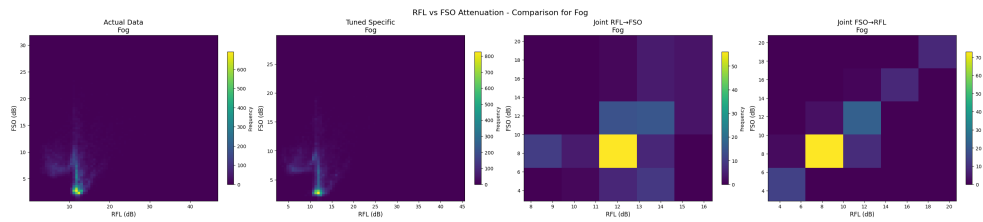


Figure 30: RF vs FSO Attenuation Heatmaps under Fog Weather

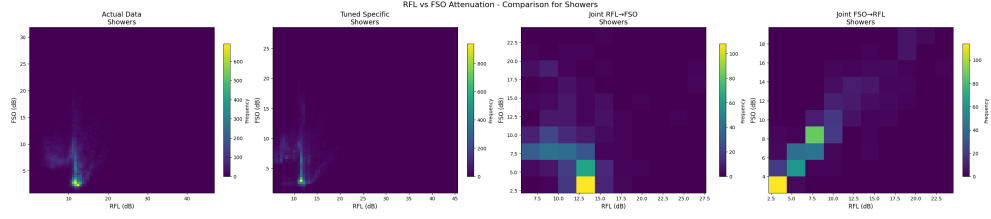


Figure 31: RF vs FSO Attenuation Heatmaps under Showers Weather

4.5 Entropy and Correlation Analysis

To further interpret how the models capture cross-channel relationships and uncertainty, we analysed two important metrics:

- Pearson Correlation Coefficient between RF and FSO attenuation
- Normalised Entropy Ratio I_{eo}/H_{eo}

Figure 32 shows both metrics across weather conditions for actual data, tuned specific model, and the two joint models. The actual data exhibits strong positive correlation under fog and snow, and a moderate to low correlation under conditions like clear skies and dust storms. Joint models attempt to reconstruct this pattern, with FSO→RF joint models typically capturing stronger correlations.

The entropy ratio provides insights into the information learned by the model. Higher values suggest better modelling of uncertainty. Across most conditions, joint models show an improvement over the tuned specific model in I_{eo}/H_{eo} , particularly under fog and drizzle. However, the overall consistency of the tuned specific model in maintaining stable correlation and entropy trends further supports its use as the base model.

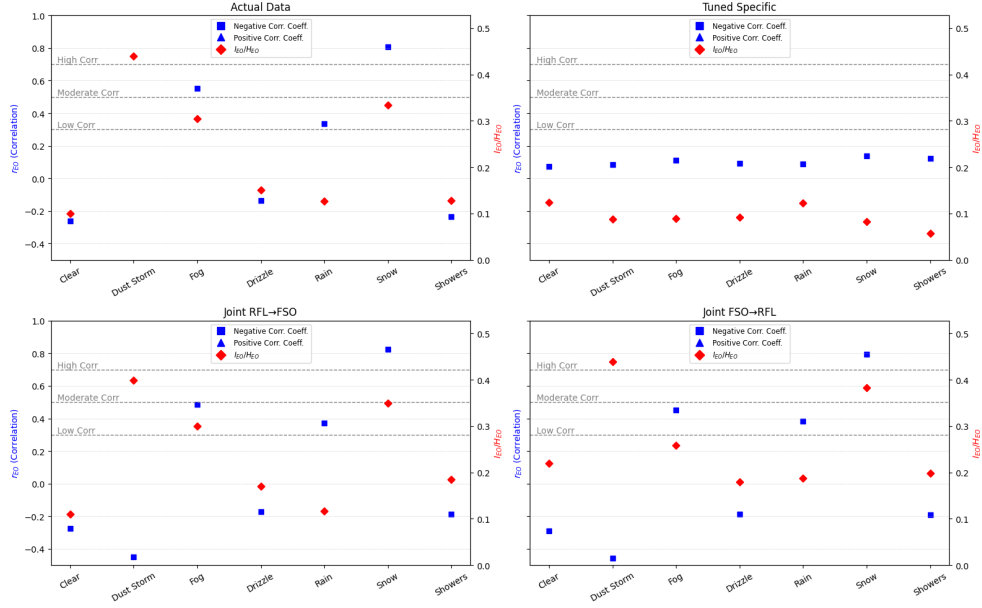


Figure 32: Comparison of Pearson Correlation and Normalised Entropy Ratio I_{eo}/H_{eo} across weather conditions (Actual, Tuned Specific, and Joint Models)

4.6 Summary

- Specific models outperformed generic models in terms of RMSE and provided better condition-specific performance.
- They were more consistent across weather types and particularly better under fog, rain, and snow.
- The tuned specific model served as a stronger base for joint modelling.
- Visual heatmaps confirmed that tuned specific models and joint models captured realistic attenuation patterns.
- Entropy and correlation analysis further highlighted the information-capturing ability and reliability of the tuned specific model.
- Visual comparison using bar and line plots further supports the decision to use specific models as the optimal baseline.

5 Conclusion

This study investigated the prediction of signal attenuation in hybrid RF/FSO communication systems under varying weather conditions using machine learning models. Leveraging a weather-tagged synthetic dataset, we modelled the relationships between environmental features (e.g., rain intensity, visibility, humidity) and channel attenuation using Random Forest regressors.

We implemented and compared three modelling strategies: specific models trained per weather condition, a generic model using all data, and joint models where one channel's predicted attenuation was used to inform the other. Performance was evaluated using RMSE and R^2 , supported by visual heatmaps and information-theoretic metrics such as Pearson correlation and entropy ratio.

Results showed that specific models consistently outperformed the generic model across nearly all conditions, especially for FSO attenuation during fog, drizzle, and snow. These models were thus selected for hyperparameter tuning and served as the foundation for joint modelling. The tuned specific models maintained strong predictive accuracy and replicated the spatial patterns of real attenuation more closely. Joint models further improved correlation and entropy scores in certain cases, highlighting cross-channel dependencies.

Visual comparisons confirmed the tuned specific model's robustness, and entropy analysis validated its ability to retain useful uncertainty patterns across diverse weather conditions. These findings suggest that weather-aware modelling is essential for hybrid systems, and that tuned, condition-specific models offer the best trade-off between accuracy and interpretability.

Overall, this research demonstrates the potential of machine learning models—especially tuned, weather-specific Random Forests—in building adaptive, resilient hybrid RF/FSO systems. Future work could integrate real-time weather feeds, investigate deep learning architectures, and explore deployment strategies in operational communication networks.

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Lastly, I also want to acknowledge that I took help with the Chat-GPT to get the syntax to create necessary format in the Latex file. Likewise, I also took help from it to create lineplots, heatmaps and correlation plot. Moreover, I also used it to understand some of the topics in deeper level and to understand the research papers used.

Appendices

GitHub Repository

The code and analysis for this project are available in the following publicly accessible GitHub repository:

GitHub Repository: Data Science Research Project

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